

EXERCISE 1C

Questions 1, 2 and 3 involve completing a weekly wage-sheet using a spreadsheet. For each of these questions the following conditions apply:

- The first 8 hours worked on any weekday are paid at the standard hourly rate
- Overtime on weekdays: first two hours, time and a half, then any extra hours double time
- Saturday hours all paid at time and a half
- Sunday hours all paid at double time
- Tax at 30c in the \$ of gross wage
- Medicare levy 2% of gross wage
- Union Fees 1% of gross wage

1. The spreadsheet below shows the hours worked by Donald Parker over the past week. Create this spreadsheet yourself and complete it using the conditions given above.

	A	B	C	D	E
1	WEEKLY WAGES SHEET				
2	Employee: Donald Parker		Standard Rate of Pay: \$		19.44
3		Total Hours	Hours Worked At		
4	Day	Worked	Standard Time	Time & Half	Double Time
5	Monday				
6	Tuesday	7			
7	Wednesday	7			
8	Thursday	7			
9	Friday	11			
10	Saturday	4			
11	Sunday	3			
12	TOTAL				
13				Bonus	
14				Travel Allowance	
15				GROSS WEEKLY PAY	
16			DEDUCTIONS	Tax	
17				Medicare	
18				Union	
19				TOTAL DEDUCTIONS	
20				WEEKLY NET PAY	
21	Pay Officer:		Employee Signature:		
22			Date:		

- (a) What formula needs to be entered in cell B12? On entering the formula what did your spreadsheet show?
- (b) What formulae need to be entered in the following cells:
 (i) Cell C12 (iii) Cell D12 (iv) Cell E12
- (c) What formula needs to be entered in cell E15? On entering the formula what did your spreadsheet show?
- (d) What formula needs to be entered in cell E16? On entering the formula what did your spreadsheet show?
- (e) What formula needs to be entered in cell E17? On entering the formula what did your spreadsheet show?
- (f) What formula needs to be entered in cell E18? On entering the formula what did your spreadsheet show?
- (g) What formula needs to be entered in cell E19? On entering the formula what did your spreadsheet show?
- (h) What formula needs to be entered in cell E20? On entering the formula what did your spreadsheet show?

2. The spreadsheet below shows the hours worked by Sally Green including travel allowance of \$107.25 over the past week.

Create this spreadsheet yourself and complete it using the given conditions.

	A	B	C	D	E
1	WEEKLY WAGES SHEET				
2	Employee: Sally Green		Standard Rate of Pay: \$ 12.80		
3	Day	Total Hours Worked	Hours Worked At		
4			Standard Time	Time & Half	Double Time
5	Monday	5			
6	Tuesday	8			
7	Wednesday	9			
8	Thursday	11			
9	Friday	6			
10	Saturday	2			
11	Sunday				
12	TOTAL				
13	Bonus				
14	Travel Allowance				107.25
15	GROSS WEEKLY PAY				
16	DEDUCTIONS		Tax		
17			Medicare		
18			Union		
19	TOTAL DEDUCTIONS				
20	WEEKLY NET PAY				
21	Pay Officer:		Employee Signature:		
22			Date:		

- (a) What formula needs to be entered in cell B12? On entering the formula what did your spreadsheet show?
- (b) What formulae need to be entered in the following cells:
 (i) Cell C12 (iii) Cell D12 (iv) Cell E12
- (c) What formula needs to be entered in cell E15? On entering the formula what did your spreadsheet show?
- (d) What formula needs to be entered in cell E16? On entering the formula what did your spreadsheet show?
- (e) What formula needs to be entered in cell E17? On entering the formula what did your spreadsheet show?
- (f) What formula needs to be entered in cell E18? On entering the formula what did your spreadsheet show?
- (g) What formula needs to be entered in cell E19? On entering the formula what did your spreadsheet show?
- (h) What formula needs to be entered in cell E20? On entering the formula what did your spreadsheet show?

3. The spreadsheet below shows the hours worked by Brian Adams over the past week including a bonus of \$200 and travel allowance of \$98.55. Create this spreadsheet yourself and complete it using the given conditions.

	A	B	C	D	E
1	WEEKLY WAGES SHEET				
2	Employee: Brian Adams		Standard Rate of Pay: \$		23.10
3		Total Hours	Hours Worked At		
4	Day	Worked	Standard Time	Time & Half	Double Time
5	Monday	9			
6	Tuesday	8			
7	Wednesday	8			
8	Thursday	10			
9	Friday	12			
10	Saturday	8			
11	Sunday	4			
12	TOTAL				
13				Bonus	200.00
14				Travel Allowance	98.55
15				GROSS WEEKLY PAY	
16			DEDUCTIONS	Tax	
17				Medicare	
18				Union	
19				TOTAL DEDUCTIONS	
20				WEEKLY NET PAY	
21	Pay Officer:		Employee Signature:		
22			Date:		

- What formula needs to be entered in cell B12? On entering the formula what did your spreadsheet show?
- What formulae need to be entered in the following cells:
 - Cell C12
 - Cell D12
 - Cell E12
- What formula needs to be entered in cell E15? On entering the formula what did your spreadsheet show?
- What formula needs to be entered in cell E16? On entering the formula what did your spreadsheet show?
- What formula needs to be entered in cell E17? On entering the formula what did your spreadsheet show?
- What formula needs to be entered in cell E18? On entering the formula what did your spreadsheet show?
- What formula needs to be entered in cell E19? On entering the formula what did your spreadsheet show?
- What formula needs to be entered in cell E20? On entering the formula what did your spreadsheet show?
- If the tax rate was increased to 35 cents in the dollar how would this affect Brian's take home pay? State this pay.
- If Brian had only worked 8 hours on Friday, how would this affect his weekly pay? State this pay.

4. The spreadsheet shows a section of the weekly wages summary sheet of a supermarket store. The employees are paid time and a half on Saturday and double time on Sundays.

	A	B	C	D	E	F
1	WEEKLY WAGES SUMMARY SHEET					
2	Employee	Pay Rate (\$/hour)	Hours worked Weekdays	Hours worked Saturday	Hours worked Sunday	Weekly Pay
3	T. Adams	18.40	30	4	5	
4	S. Currie	21.45	34	5	3	
5	F. Egan	16.88	6	6	4	
6	M. Johns	19.34	28	8	2	
7	B. Marshall	23.56	33	5	5	

- (a) Calculate how much T. Adams earned during this week.
- (b) Without performing any calculations determine which of the given employees earned the least during this week. Justify your answer.
- (c) Using cell references, state the formula that can be used to calculate the weekly pay of T. Adams.
- (d) Using cell references, state the formula that can be used to calculate the weekly pay of F. Egan.
- (e) Create this spreadsheet on your computer or calculator and list the weekly pay for each of the given employees.

5. The spreadsheet shows a section of the weekly wages summary sheet for Freddie's Fast Food store. Fast food employees are paid 25% above their hourly rate for hours worked on Saturday, 50% above their hourly rate for hours worked on Sunday and 150% above their hourly rate for hours worked on Public Holidays.

	A	B	C	D	E	F	G
1	WEEKLY WAGES SUMMARY SHEET						
2	Employee	Hourly Rate(\$)	Hours worked Weekdays	Hours worked Saturday	Hours worked Sunday	Public Holiday Hours	Weekly Pay
3	Allan	18.52	25	5	4	4	875.07
4	Jake	19.64	32	4	3	0	
5	Mavis	19.64	29	7	2	0	
6	Niamh	19.64	33	6	4	6	
7	George	19.94	28	8	3	4	
8	Harriet	20.18	26	0	4	5	

- (a) How much did Niamh earn this week?
- (b) Using cell references, state the formula that can be used to calculate Allan's weekly pay.
- (c) Create this spreadsheet on your computer or calculator and list the weekly pay for each of the given employees.

11. Alan Smith and his wife Grace have a combine after tax income of \$76400. They are buying a house and have monthly mortgage repayments of \$2028. Alan and Grace have tabled their yearly expenses, shown below, in order to create a sensible weekly budget.

FIXED EXPENSES		VARIABLE EXPENSES		DISCRETIONARY EXPENSES	
Home loan repayment	\$24 336	Food	\$9000	Entertainment	\$2600
Car loan repayment	8000	Clothing	2500	Holidays	2500
House Insurance	1250	Electricity and gas	3600	Gifts	800
Health Insurance	1980	Car running expenses	5800	Sundries	3500
Car Insurance	980	Telephone and internet	1860		
Car Registration	680	Fares(bus, train, taxi)	650		
Rates	3450				

- (a) What is meant by the terms Fixed, Variable and Discretionary expenses?

	A	B	C
1	SMITH FAMILY BUDGET		
2			
3	Income(net)	Annual	Weekly
4	Alan	53000	
5	Grace	23400	
6	Total	76400	
7			
8	Expenses		
9	Fixed		
10	Home loan	24336	
11	Car loan	8000	
12	House insurance	1250	
13	Health insurance	1980	
14	Car insurance	980	
15	Car registration	680	
16	Rates	3450	
17	Total		
18			
19	Variable		
20	Food	9000	
21	Clothing	2500	
22	Elect & Gas	3600	
23	Car Expenses	5800	
24	Phone & Internet	1860	
25	Fares	650	
26	Total		
27			
28	Discretionary		
29	Entertainment	2600	
30	Holidays	2500	
31	Gifts	800	
32	Sundries	3500	
33	Total		
34			
35	SURPLUS		
36			

On your computer or calculator create a spreadsheet for the Smith family as shown opposite.

- (b) Find the total fixed, variable and discretionary expenses.
- (c) If the Smith family stick to this yearly budget, how much will they save during the year?
- (d) What formula must be entered in:
 (i) cell B17?
 (ii) cell B35
 (iii) cell C6?
 (iv) cell C20?
- (e) Explain how the cells C4 to C35 can be filled.
- (f) How much must Alan and Grace put aside each week to cover their annual fixed expenses.
- (g) How much on average can the family spend weekly on:
 (i) food? (nearest 5 dollars)
 (ii) clothing? (nearest 5 dollars)
 (iii) entertainment?

12. Catherine works part-time at a supermarket and part time at a café to support herself while she undertakes studies at university.
 Catherine's income changes from week to week but she expects weekly take home pay of \$240 from the supermarket and \$210 from the café each week.
 To keep track of her spending Catherine prepares a weekly budget, she budgets \$140 for rent, \$65 for car repayments, \$48 for car insurance and \$40 for petrol and car maintenance.
 She sets aside \$55 for food, \$15 for her mobile phone, \$25 for entertainment, sundries \$20, health insurance \$30.
 Any surplus money Catherine deposits into a savings account.
 (a) Expenditure can be classified as being fixed, variable or discretionary. Explain the meaning of fixed, variable and discretionary expenditure.

- (b) Set up a spreadsheet like the one below and name it 'Catherine's Weekly Budget Sheet'.

	A	B	C	D
1	Catherine's Weekly Budget Sheet			
2				
3		Budgeted Amount	Actual Amount	
4	Income			
5	Supermarket			
6	Café			
7	Other			
8	Total			
9				
10	Fixed Expenses			
11	Rent			
12	Car repayment			
13	Car insurance			
14	Health cover			
15	Total			
16				
17	Variable expenses			
18	Petrol and maintenance			
19	Food			
20	Mobile phone			
21	Total			
22				
23	Discretionary Expenses			
24	Entertainment			
25	Sundries			
26	Total			
27				
28	Summary			
29	Income			
30	Fixed expenses			
31	Variable expenses			
32	Discretionary expenses			
33	Income - Expenses			
34				

- (c) List the formulae that need to be entered in the following cells:
- (i) cell B8
 - (ii) cell B15
 - (iii) cell B21
 - (iv) cell B33

- (d) How much must Catherine set aside each week to cover her fixed expenses?
- (e) What percentage of her weekly budget is set aside for car expenses?

During this past week in addition to her expected supermarket and café earnings, Catherine earned \$85 tutoring mathematics. Catherine spent \$70 on petrol and car maintenance because she had to replace the car battery. She spent \$52 on food, phone \$15, \$10 on stockings and \$12 on stationary. Catherine did not spend any money on entertainment as she was too busy with work, tutoring and preparing a university assignment and tutorial. Enter Catherine's actual income and expenses onto the spreadsheet prepared for (b).

- (f) List the formulae that need to be entered in the following cells:
- (i) cell C8
 - (ii) cell C33

- (g) Which categories of expenses changed, and by how much?

- (h) What is the difference between Catherine's planned expenses and her actual expenses?

- (i) Where did Catherine spend more than she had planned?

- (j) Where did Catherine spend less than she had planned?

- (k) What percentage of her income did Catherine spend on her car this week?

- (l) Calculate Catherine's monthly expenditure to the nearest dollar on:
- (i) rent,
 - (ii) health insurance.

- (m) How much money did Catherine add to her savings this week?

- (n) Discuss how Catherine can increase her weekly savings.

13. Craig lives at home with his parents and his weekly take home pay as an employee at a fast food outlet is \$438. His expenses are as follows: Board \$110 per week; monthly car repayments \$364; car insurance \$1612 p.a.; petrol costs \$40 each week; lunches \$50 each week; gym membership \$195 per quarter and entertainment \$40 each week.

(a) Classify Craig's expenses using the following table.

Fixed Expenses	Variable Expenses	Discretionary Expenses

(b) Making use of the expenses classification from (a) prepare a weekly budget for Craig using the spreadsheet below.

	A	B	C	D
1	CRAIG'S WEEKLY BUDGET			
2				
3	Income			
4				
5		Payment Period	Cost per Period	Weekly Cost
6	Fixed Expenses			
7				
8				
9				
10				
11	Variable Expenses			
12				
13				
14				
15	Discretionary Expenses			
16				
17				
18				
19				
20	Total Expenses			
21				
22	Surplus/Deficit			
23				

(c) What formulae must be entered in the following cells:

(i) cell D8

(ii) cell D9

(iii) cell D17

(iv) cell D20

(v) cell D22

(d) If Craig sticks to his budget, how much of his take home pay is he able to save each week?

(e) What percentage of Craig's take home pay is spent on car expenses?

(f) Craig wishes to go on an overseas holiday and estimates that he will need \$3000 for fares, accommodation and spending money. How long will it take Craig to save up for his holiday if he sticks to his budget?

(g) Craig decides to save half of his entertainment allowance in order to accumulate \$3000 at a faster rate. How much earlier will he be able to go on his overseas holiday as a result of this saving?